Shahariyar Maraz

119/A,Shershah Suri Road, Mohammadpur, Dhaka-1207

(+880) 1521438476 | shahariyarrme@gmail.com | GitHub | LinkedIn

Personal Profile

My technical proficiency covers a wide spectrum of microcontrollers and embedded platforms, including the ESP32-S2, Raspberry Pi, Arduino UNO (ATmega328P), Arduino Mega (ATmega2560), Arduino Due (Atmel SAM3X8E ARM Cortex-M3), and Arduino Nano (ATmega328). I have a strong foundation in embedded system design, firmware development, FreeRTOS, signal conditioning circuit design, data analysis, and communication protocols (UART, I^2C, SPI, SDI-12, etc.), as well as debugging and testing. In addition, I am experienced with Wi-Fi and GSM-based systems, including hands-on expertise with AT command–driven GSM modules. Currently, I am directing my efforts towards the creation of wearable medical assistive devices

Education

Masters of Science in Robotics and Mechatronics Engineering — *Dhaka,Bangladesh*
University of Dhaka — *2023-2024*
CGPA: 3.65 out of 4.0

Bachelor of Science in Robotics and Mechatronics Engineering — *Dhaka,Bangladesh*
University of Dhaka — *2017 - 2021*
CGPA: 3.08 out of 4.0

HSC in Science — *Dhaka,Bangladesh*
ST. Joseph Higher Secondary School — *2014 - 2015*
GPA: 5.00 out of 5.00

SSC in Science — *Kurigram,Bangladesh*
Kurigram Govt. High School — *2012 - 2013*
GPA: 5.00 out of 5.00

Working Experience

Hardware Engineer — *Dhaka, Bangladesh*
Datasoft Manufacturing & Assembly Limited — *May 2025 - Present*

- **IoT Smart Battery Management System:** Developed an embedded system that monitors battery health using multiple sensors and transmits data over the mobile network, It increases sampling during cranking, stores data when the signal drops, automatically recovers the modem and retransmits cached data, and provides LED indicators for technician diagnostics.
- **IoT Water Level Monitoring System of River:** Designed and developed a microcontroller-based data logger for river water level monitoring using a radar stage sensor. Implemented SDI-12 sensor integration, LOS/AOS data acquisition, real-time data processing, SD card logging, GSM/MQTT telemetry, and power management for reliable field deployment.
- **Water Quality Monitor:** Designed a data logging system for hydrological applications to measure pH, total dissolved solids (TDS), dissolved oxygen (DO), ammonia (NH_3) concentration, water velocity, and water level stage.

Research Assistant and Lab Manager — *Dhaka,Bangladesh*
MAIM Lab, Funded by Wellcome Leap (In Utero), California, USA , and University Grand Commission of Bangladesh — *October 2023-Present*
Collaborator: University College Dublin, Ireland

- Translation of a Wearable Fetal Movement Monitor towards Stillbirth Prevention (Wellcome Leap Funded).
- Mechanomyography-Based Wearable Headset for Eye-blink Gesture Recognition (UGC Funded).

Jr. Embedded System Engineer *Dhaka, Bangladesh*
***Gizantech* [Work Repositories]** *January 2025 - April 2025*

- Acquired professional PCB design skills and developed pcb modules including a GSM interface and multiple DC-DC converters.

- Contributing in the firmware and PCB Design of Fusion_SL_Booster_V1.0.1 Project.

Ongoing Research Project

Design and Development of a Mechanomyography-Based Wearable Headset for Eye-blink Gesture Recognition *Dhaka, Bangladesh*
Funded by University Grant Commission of Bangladesh *2023-2024*
MSc. Thesis, Resources: [Presentation] [Hardware & Software], Advisor: Dr. Abhishek Kumar Ghosh;

- Designing an Assistive Wearable Headband: This initiative aims to develop a headband utilizing Mechanomyography (MMG) technology to benefit individuals with motor impairment.
- Integration of Multi-Sensor System: The headband incorporates piezoelectric and accelerometer sensors to capture eyelid-muscle movements, along with a data acquisition (DAQ) system integrated with an inertial measurement unit (IMU) for head tracking.
- Signal Processing: Utilizing Discrete Fourier Transform (DFT) techniques, we analyze the sensor signal dynamics and develop a customized signal conditioning circuit on a PCB board to fulfill specific criteria.
- Development of Firmware: In order to ensure real-time processing capabilities, we devised an optimized firmware for the ESP32-S2, incorporating timers, semaphores and circular buffer.
- Eyeblink Gesture Design: Subsequent to gathering data from participants, we will perform analysis to focus on the design of distinct eyeblink gestures.

Programming/Design Skills

Programming Languages/Platforms:	C/C++, Arduino/ESP, FreeRTOS **[Work Samples]**, Python **[Work Samples]**, HTML/CSS, LaTeX, MATLAB
Soft Skills:	GraphPad Prism **[Work Samples]**, Proteus **[Work Samples]**, MS Office
PCB Design:	EasyEDA **[Work Samples]**, Kicad **[Work Samples]**
3D Design:	Solidworks **[Work Samples]**
2D Design:	Adobe Illustrator **[Work Samples]**

Research Experience

Design and Control of Automated Medical Bed Systems Using IoT *Dhaka, Bangladesh*
BSc. Project, Resources: [Thesis book] [Presentation] [Demo Video] *2020-2021*

- Addressing the challenges of transporting patients within medical facilities, we have designed and developed a web server-based automated system. By inputting the destination and time information into the server, the medical bed will autonomously navigates to the specified location.
- The medical bed is capable of receiving patients from the entry gate of the medical and When the bed arrives at the gate, it will automatically fold down to accommodate the patient's transfer into the bed.
- The web server enables us to remotely track the patient's location and verify the vacancy status of medical cabins, particularly in emergency situations.
- During pandemic situations, when low human intervention is essential for running medical facilities, this system proves to be highly advantageous.
- **Physical Demonstration:** We designed a prototype of a medical bed constructed with PVC. The design includes a linear actuation system, enabling the bed to be integrated with robotic mechanisms. The prototype is assembled with a stepper motor to achieve the precise linear movements of bed, gear motor to support the actuation of the overall medical bed system, a camera for navigating the QR path and recognizing the room number, motor controller to coordinate the motor's actions, a Raspberry Pi MCU and battery to ensure the power.

University Projects

Linear actuator-based system control by IoT *Dhaka,Bangladesh*
Resources: [Code Base] *2023*

- Feedback Position Monitoring: Developed an IoT-based control system for linear actuators, enabling real-time motor position feedback on a web interface.
- Bidirectional Communication: Implemented bidirectional communication and cloud-based data visualization for real-time actuator control(Extend & Retract) and performance monitoring, enhancing precision, user interactivity, and system responsiveness.
- To build this system we have used Arduino Giga R1 WiFi which is based on STM32H747XI dual Cortex®-M7+M4 32bit low power Arm® MCU, feedback controlled linear actuator and L298N controller.

Fire Extinguishers Robot
Dhaka,Bangladesh

Resources: [Presentation][Prototype][Code] *2019*

- Solution Integration: By leveraging advancements in these fields, the bot can autonomously detect fires, navigate indoor environments, and extinguish fires effectively, improving overall safety.
- Algorithm for Fire Detection: Through temperature and gas sensors, the bot detects impending fires, signaling alerts in advance. It then autonomously seeks out the fire and moves towards it. With a built-in water tank (exchangeable with CO2), it deploys a nozzle to extinguish the flames.
- **Sensor Integration:** The bot integrates Flame Sensor Module, DHT22 Temperature and Humidity Sensor, and MQ2 Gas Sensor to detect anomalies efficiently. Flame Sensor Module identifies flames, triggering the bot's protocol. DHT22 provides temperature and humidity data, aiding in early fire detection. MQ2 Gas Sensor detects combustible gases, enhancing fire risk assessment.
- **Hardware Fabrication:** Designed and developed with four feedback-controlled motors for precise speed and location management, the bot's hardware includes a dedicated pump motor facilitating water transfer from the reservoir. An integral precision nozzle ensures accurate and high-speed water discharge and a Arduino Mega MCU for controlling the overall system.

Self Solving Eight Puzzle
Dhaka,Bangladesh

Resources: [Demo Video] *2019*

- Solving Eight Puzzle with utilizing AI: Designed and executed the operation of an eight puzzle solver program based on the A* search algorithm.
- Uneven Tiles Heuristic: This heuristic function looks at the number of tiles that's are incorrectly placed, aiming to lead the search to reached at the goal state.
- **Fabrication of Designed Model:** We designed and assembled an embedded board employing Seven-segment display, BCD to seven-segment decoder (7447), Arduino UNO, and Vero board.
- **Physical Demonstration:** An integrated board featuring nine discrete displays, each showcasing a digit from 1 to 8, while 0 signifies an unoccupied position. Its function involves reordering an arbitrary sequence of digits between 1 and 8 by exchanging places with 0 to achieve puzzle resolution.

Cycloconverter
Dhaka,Bangladesh

Resources: [Simulation][Prototype] *2018*

- Single Phase Cycloconverter: The aim of the project was the design and development of a Cycloconverter system intended to precise regulations of single phase AC motor's speed by transforming single phase alternating current (AC) from one frequency to another.
- Application of Cycloconverter: Applications in various industries, including speed control of AC motors, frequency conversion for power transmission systems, adjustable speed drives for industrial processes, grid stabilization and power quality improvement.
- **Simulation Analysis:** Utilizing Proteus, we conducted extensive modeling and simulation of the cycloconverter system to assess its performance under different operational parameters.
- **Physical Demonstration:** The fabrication of a protective housing for the essential components of cycloconverter was designed using PVC board material. Concurrently, a set of essential components including transformer, diodes, relays, and an Arduino Uno microcontroller were assembled into the case and an algorithmic approach was devised to regulate the frequency modulation of the AC signal

Sumobot
Dhaka,Bangladesh

Resources: [Prototype] *2017*

- Sumobot refers to a type of robot designed for competitive sumo wrestling matches. These robots are typically small, agile, and equipped with sensors to detect the edge of the arena or the opponent. The objective of sumobot competitions is to force the opponent's robot out of the ring within a specified time limit.
- **Sumobot Design and Construction:** Our approach to designing the sumobot involved fabricating a PVC chassis with dual inclined sides. These incline sides were strategically engineered, which often led to difficulties in accurately locating the sumobot's position. The assembly integrated a set of components including ultrasonic sound sensors, IR sensors, high-torque motors, a motor controller, battery, and an Arduino Uno microcontroller, all housed within the PVC case.
- **Competition Experience:** At the Intra department sumobot championship-2017 organized by the Robotics and Mechatronics Engineering department of the University of Dhaka, our team clinched the prestigious title of first runner-up.

Volunteering Experience

Dhaka University Photographic Society (DUPS)
Dhaka,Bangladesh

Executive Member *2017-2018*

- **Workshop and Training:** Successfully completed a basic photography course organised by DUPS and engaged in multiple workshops on different aspects of photography, including street photography.
- **Photo Exhibition Volunteering:** Coordinated various photographic festivals at DUPS, including the Annual Photography Exhibitions, National Photography Festival, and the May Day Photography Exhibition. Also, Took the initiative to organize the club's weekly meeting and photo walk.

Scholarship & Funding Information

2024 & 2023 UGC-Funded Research Grant: Received recognition for an exceptional research idea in the M.Sc. study titled "Design and Development of a Mechanomyography-Based Wearable Headset for Eye-blink Gesture Recognition".

2023 & 2022 IFIC Bank Scholarship: Awarded for excellent research potential in the M.Sc. research (Design and Development of a Mechanomyography-Based Wearable Headset for Eye-blink Gesture Recognition).

Reference

Dr. Abhishek Kumar Ghosh

Assistant professor

Department of Robotics and Mechatronics Engineering, University of Dhaka

PhD in Mechanical Engineering - Imperial College London, United Kingdom

Email: abhishek.rme@du.ac.bd

Google Scholar: Dr. Abhishek Kumar Ghosh

Mobile: +8801602908408